

First, the polyhedron needs to be written in standard form. In order to do so, we need to transform the constraints into equality constraints by introducing the slack variables x_3 and x_4 for the first and second equation, respectively. The constraints are then written as follows:

$$\begin{aligned} -x_1 + x_2 + x_3 &= 2, \\ x_1 + x_2 + x_4 &= 4, \\ x_1, x_2, x_3, x_4 &\geq 0. \end{aligned}$$

Given a feasible point of the polyhedron, it is possible to analyze if a direction is feasible by checking the following two conditions:

- $Ad = 0$, and
- $d_i \geq 0$ when $x_i = 0$,

where

$$A = \begin{pmatrix} -1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{pmatrix}, x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}, d = \begin{pmatrix} d_1 \\ d_2 \\ d_3 \\ d_4 \end{pmatrix}.$$

Let's verify if these conditions are satisfied for the four cases.

$$1. \tilde{x}_a = \begin{pmatrix} 0 \\ 2 \end{pmatrix}, \tilde{d}_a = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

We first need to compute the values of the slack variables x_3 and x_4 and we obtain $x_3 = 0$ and $x_4 = 2$. Therefore, we have

$$x_a = \begin{pmatrix} 0 \\ 2 \\ 0 \\ 2 \end{pmatrix},$$

which is indeed a feasible point. The direction in $\mathbb{R}^4$ is

$$d_a = \begin{pmatrix} 1 \\ 0 \\ d_3 \\ d_4 \end{pmatrix}.$$

In order to satisfy the first condition, we must have

$$Ad_a = \begin{pmatrix} -1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ d_3 \\ d_4 \end{pmatrix} = \begin{pmatrix} -1 + d_3 \\ 1 + d_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

which is verified by setting $d_3 = 1$ and $d_4 = -1$. Therefore, the direction

$$d_a = \begin{pmatrix} 1 \\ 0 \\ 1 \\ -1 \end{pmatrix}$$

satisfies the first condition. For the second condition, we look at the entries 1 and 3 of d_a , because $(x_a)_1 = (x_a)_3 = 0$ and the only zero entries. As $(d_a)_1 = 1$ and $(d_a)_3 = 1$, both entries of the direction are non negative and we conclude that the direction is feasible at the given point.

$$2. \tilde{x}_b = \begin{pmatrix} 0 \\ 2 \end{pmatrix}, \tilde{d}_b = \begin{pmatrix} -1 \\ 0 \end{pmatrix}.$$

In this case, it is immediate to see that the direction is not feasible, as the second condition is not verified for the first entry: $(\tilde{d}_b)_1 = -1 < 0$ and $(\tilde{x}_b)_1 = 0$.

$$3. \tilde{x}_c = \begin{pmatrix} 2 \\ 2 \end{pmatrix}, \tilde{d}_c = \begin{pmatrix} 0 \\ 2 \end{pmatrix}.$$

We compute the values of the slack variables: $x_3 = 2, x_4 = 0$, so that

$$x_c = \begin{pmatrix} 2 \\ 2 \\ 2 \\ 0 \end{pmatrix},$$

and the direction is

$$d_c = \begin{pmatrix} 0 \\ 2 \\ d_3 \\ d_4 \end{pmatrix}.$$

For the first condition to be satisfied, we must have

$$Ad = \begin{pmatrix} -1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 2 \\ d_3 \\ d_4 \end{pmatrix} = \begin{pmatrix} 2 + d_3 \\ 2 + d_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

which is verified if $d_3 = -2$ and $d_4 = -2$. Therefore,

$$d_c = \begin{pmatrix} 0 \\ 2 \\ -2 \\ -2 \end{pmatrix}$$

satisfies the first condition. For the second condition, we need to look at the entry 4 of d_c , as $(x_c)_4 = 0$ is the only zero entry of x_c . As $(d_c)_4 = -2 < 0$, the second condition is not satisfied, and the direction is not feasible at the given point.

$$4. \tilde{x}_d = \begin{pmatrix} 3 \\ 0 \end{pmatrix}, \tilde{d}_d = \begin{pmatrix} 0 \\ 2 \end{pmatrix}$$

We compute the values of the slack variables: $x_3 = 5, x_4 = 1$, so that

$$x_d = \begin{pmatrix} 3 \\ 0 \\ 5 \\ 1 \end{pmatrix},$$

and the direction is

$$d_d = \begin{pmatrix} 0 \\ 2 \\ d_3 \\ d_4 \end{pmatrix}.$$

For the first condition to be satisfied, we must have

$$Ad_d = \begin{pmatrix} -1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 2 \\ d_3 \\ d_4 \end{pmatrix} = \begin{pmatrix} 2 + d_3 \\ 2 + d_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

from which we obtain $d_3 = -2$ and $d_4 = -2$. Therefore,

$$d_d = \begin{pmatrix} 0 \\ 2 \\ -2 \\ -2 \end{pmatrix}$$

verifies the first condition. For the second condition, we need to look at the entry 2 of d , as $x_2 = 0$ is the only zero entry of x . As $d_2 = 2 \geq 0$, the second condition is satisfied, and the direction is feasible at the given point.

The following figure presents the four studied cases.

